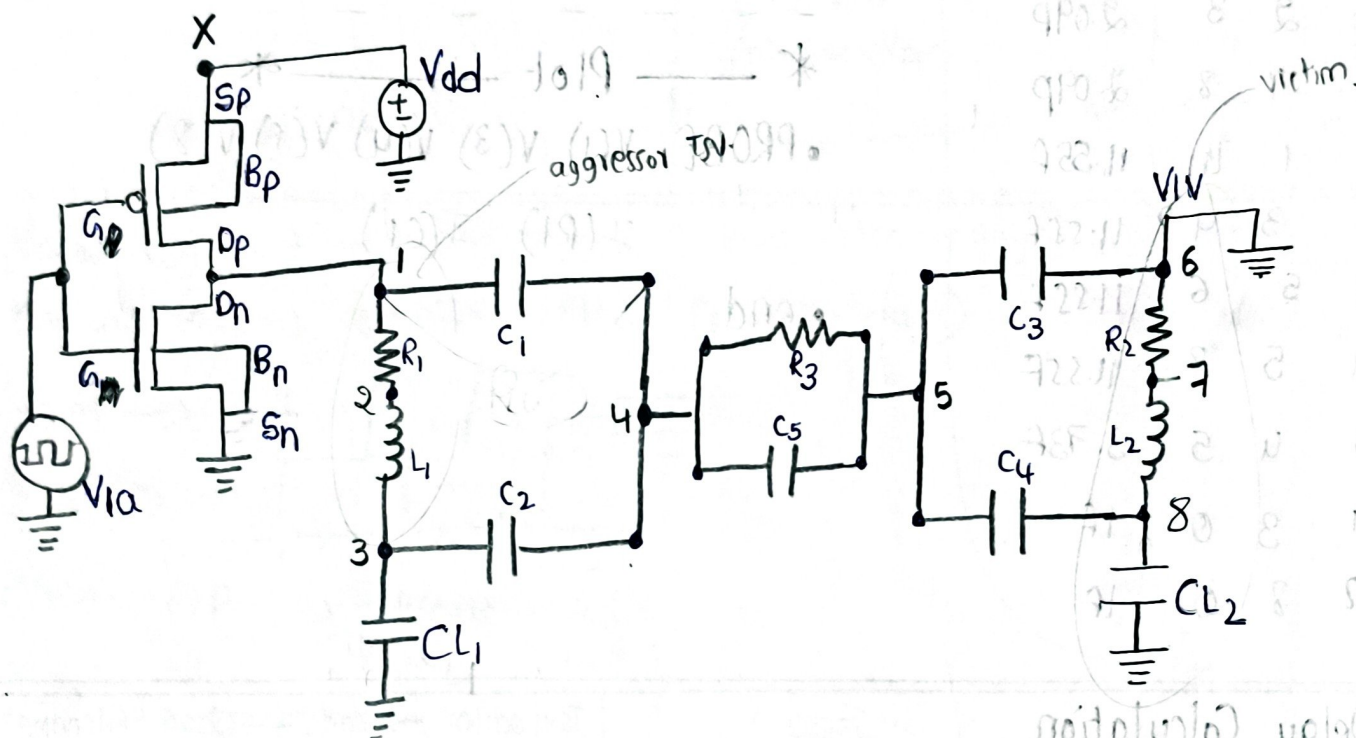


18/06/26 Day 23 kamineni Dhana Sri-keerthi

Defect Free

Circuit: "Delay" (Propagational) / "Noise"



Code:

```

temp27
include /home/vlsi/Desktop/10IP-k.d.s.k/Library/22nm_HP.pm (Technology library file copy)

```

* Supply *

```

v1a g 0 pulse (0 0.8 0 100p 100p 5n 0n)
vdd x 0 0.8
v1v 6 0 DC 1

```

* Driver *

Mp	1	G	X	X	pmos	w=88n	l=22n
Mn	1	G	0	0	nmos	w=44n	l=22n
	drain dp, dn	gate gn	source sp, sn	Body. bp, bn			

* Network *

R ₁	1	2	82.8m
R ₂	6	7	82.8m
R ₃	4	5	35uH
L ₁	2	3	2.09p
L ₂	7	8	2.09p
C ₁	1	4	11.55f
C ₂	3	4	11.55f
C ₃	5	6	11.55f
C ₄	5	8	11.55f
C ₅	4	5	3.73f
CL1	3	0	1f
CL2	8	0	1f

* Operation *

• Option post

• trans ip 100n

* measure *

* Plot *

• PROBE V(1) V(3) V(4) V(5) V(8)

I(R1) I(C1)

• end

01. Delay Calculation.

Rise time = 0.45f

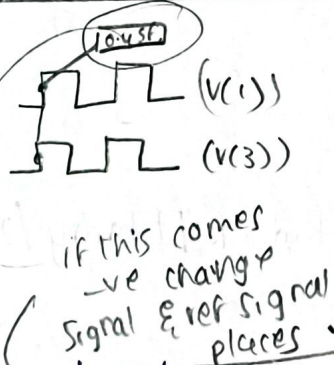
Fall time = 0.53f

propagation delay = $\frac{t_r + t_f}{2}$

$$= \frac{0.45f + 0.53f}{2}$$

$$= \frac{0.98}{2}$$

$$PD = 0.49f$$



Text editor → code → save "filename"

→ goback → open folder this file saved in

→ open in terminal → type

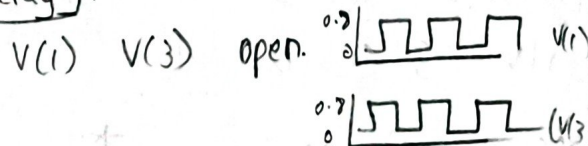
hspice filename → click enter →

it will show hspice job concluded

→ type wr in next line → enter

→ open wave form.

Delay:



(risetime)

Measure tool → Time domain(Delay)

Signal: (V(3))

Ref. sig: (V(1))

delay level

Signal at 50 percentage

refsig at 50 percentage

Target signal

edge type

o Rise

o Some

continue.

Continue

for fall time just change.

Edge type

☐ Fall
☒ samp

crosstalk glitch.

Measure b/w closest edges by default.

Measure

either

delay.

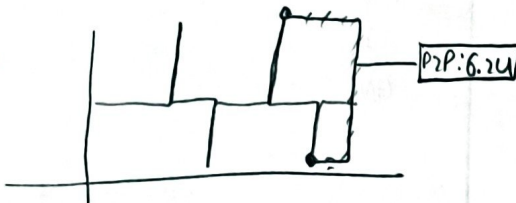
X range option

on fire waveform

Noise: (crosstalk with victim is quieted intentionally)

Measure tool → left side → level (peak to peak) → signal V(2)

→ ok →



$$\text{Noise} = \frac{P2P}{2} \Rightarrow \text{Noise} = \frac{6.24}{2}$$

$$\Rightarrow \text{Noise} = 3.1 \mu\text{V}$$

Crosstalk delay.

0:1 Crosstalk (inphase)

code (change)

* Supply *

vla g @ pulse (0 0.8 0 100p 100p sn 10n)

wd x 0 0.8

vlv y @ pulse (0 0.8 0 100p 100p sn 10n)

* driver *

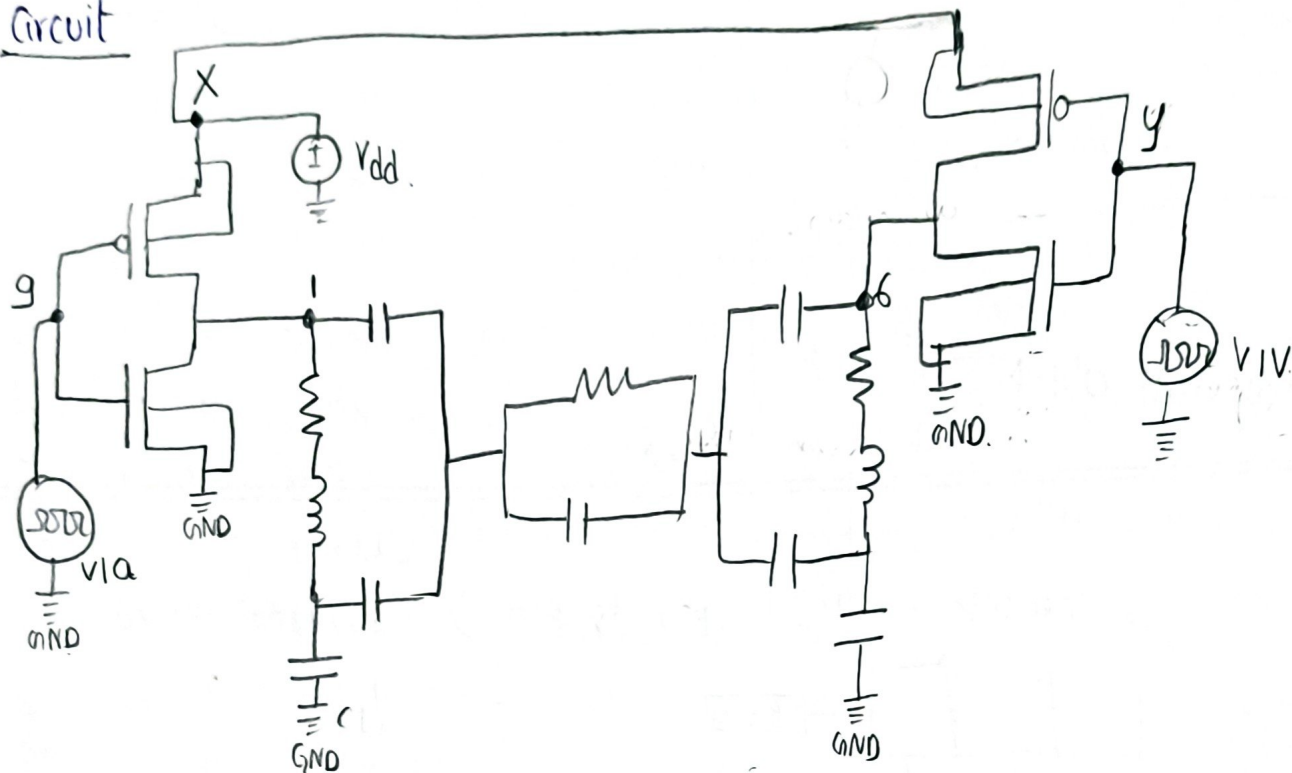
Mpa 1 G X X w=88n l=22n

Mna 1 G 0 0 w=44n l=22n

Mpv 6 y X X w=88n l=22n

Mnv 6 y 0 0 w=44n l=22n

Circuit



~~Design~~

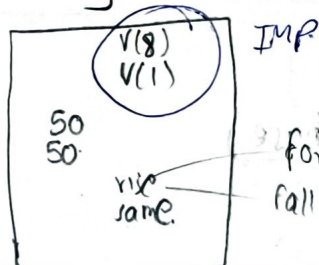
Crosstalk (same as ~~rise~~ delay but with a little circuit change (2 drivers))

$$\rightarrow t_r = 0.11f \quad t_f = 0.13f$$

$$CT(\text{inphase}) = 0.11 + 0.13f = 0.12f$$

open v(1) & v(8) in wave view [left up double click file $\rightarrow$ top level $\rightarrow$ left bottom v(1) v(2) - - -]

Measure tool $\rightarrow$ Timedomain $\rightarrow$ delay $\rightarrow$



0.2 Crosstalk outphase.

code:

** supply **

v1a g 0 pulse(0 0.8 0 100p 100p sn 10n)

Vdd x 0 0.8

v1v y 0 pulse(0.8 0 0 100p 100p sn 10n)

measure tool $\rightarrow$ timedomain $\rightarrow$

delay $\rightarrow$



$$t_r = 5.1n \quad t_f = 5.1n$$

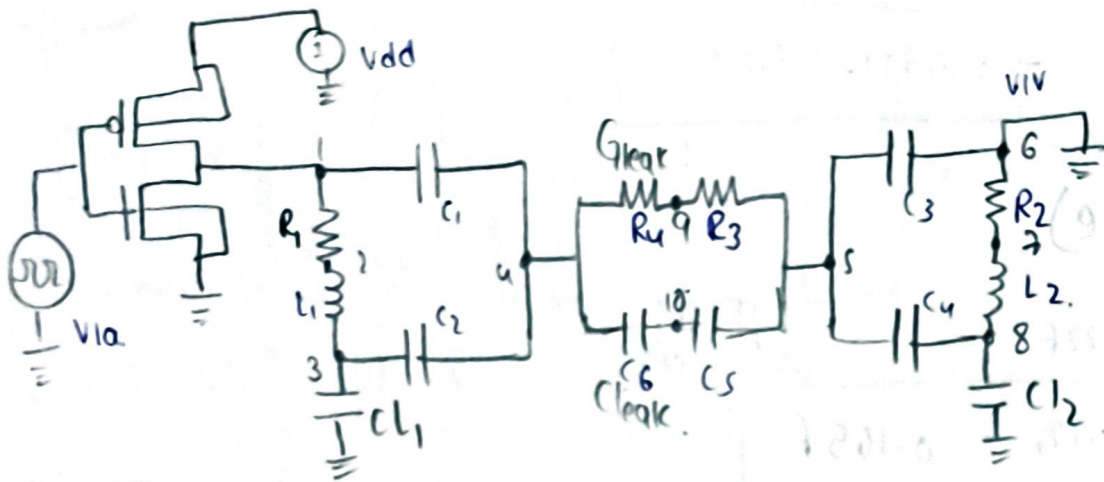
$$CT(\text{outphase}) = \frac{5.1 + 5.1}{2} = \frac{10.2}{2}$$

$$CT_{\text{out}} = 5.1n$$

8/06/26 Day23 kamineni Dhona Sri Keerthi

Defected CG [C1].[P]

oi Delay and Noise,



ii code

In net work add this)

R1 1 2 82.8m

R2 6 7 82.8m

R3 9 5 259u.

R4 4 9 -214.3 / 108.4u P

L1 2 3 12.09p

L2 7 8 2.09p

C1 1 4 11.25f

C2 3 4 11.25f

C3 5 6 11.25f

C4 5 8 11.25f

C5 10 5 2.73f

C6 4 10 64f

C11 3 0 1f

C12 8 0 1f

(2236) (2201/100) = 11.25

$$C_{ins} = \frac{C_{ox} (dep_1 / (11.25))}{C_{ox} + dep_1}$$

$$= \frac{(65f)(31.4f)}{65f + 31.4f}$$

a) $C_{ins} = 22.59f$

$$C_{ins}/2 = 11.25 - \text{value}$$

i2plot +

.PROBE V(1) V(3) V(4)
V(9) V(10) V(5)
V(7) V(8)
I(C1) I(R4)

.end

delay - cl

$$t_r = 0.59f \quad t_f = 0.51f$$

$$PD_{cl} = 0.55f$$

Noise

$$\frac{P2P}{2} = \frac{6.14 \mu}{2} = 3.07 \mu = \text{Noise.}$$

Crosstalk (inphase).

$$t_r = 0.11f \quad t_f = 0.22f$$

$$CT_{in} = \frac{0.11 + 0.22f}{2} = 0.165f$$

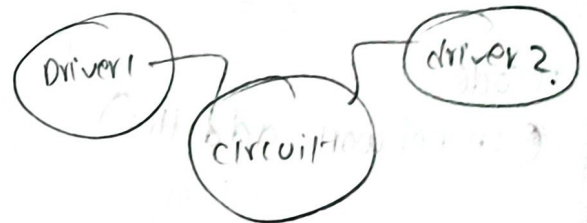
Crosstalk (outphase).

$$t_r = 5.1n \quad t_f = 5.1n$$

$$CT_{out} = 5.1n$$

→ circuit and driver add.

see files 180626 files.



→ Proposed.

(For this change R_u Co values remain all same as cl codes.)

delay - p'

$$\frac{0.59f + 0.5f}{2} = 0.545f$$

CT-in

$$\frac{(0.21 + 0.13)f}{2} = 0.17f$$

Noise

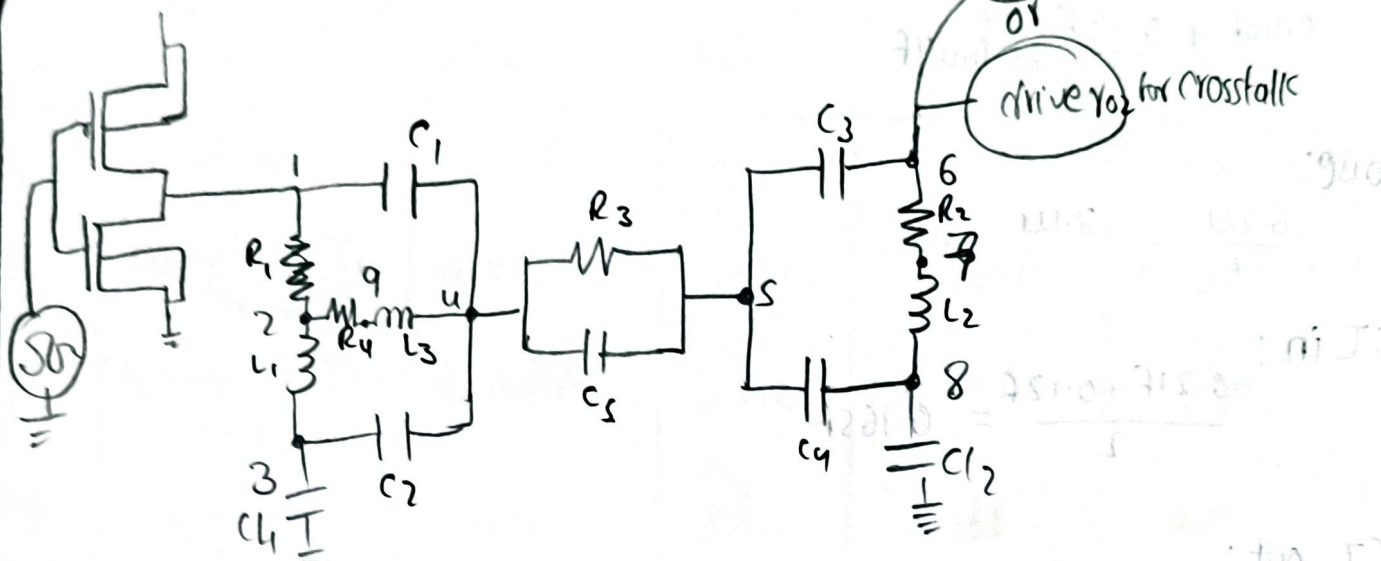
$$\frac{6.14 \mu}{2} = 3.07 \mu$$

CT-out

$$\frac{5.1n + 5.1n}{2} = 5.1n.$$

defected RL (N,P)

Circuit.



R_1	1	2	82.8m
R_2	6	7	82.8m
R_3	4	5	354μ
R_4	2	9	0.136m / 0.133m
L_1	2	3	2.09p
L_2	7	8	2.09p
L_3	9	4	1.901e-20 / 2.05e-20
C_1	1	4	11.556f
C_2	3	4	11.556f
C_3	5	6	11.556f
C_4	5	8	11.556f.
C_5	4	5	3.73f
C_{11}	3	0	1f
C_{12}	8	0	1f.

$$\begin{aligned}
 C_{ins} &= \frac{C_{ox} C_{dep}}{C_{ox} + C_{dep}} \\
 &= \frac{(65.07f)(35.84f)}{65.07f + 35.84f} \\
 &= \frac{2,332.1028}{100.91} \\
 &= 23.113f
 \end{aligned}$$

$$C_{ins} = 11.5565f$$

PROBE $V(1)$ $V(3)$ $V(9)$ $V(4)$ $V(5)$ $V(7)$ $V(8)$:
 $I(C_1)$ $I(R_4)$.

(N)

delay:

$$\frac{0.46f + 0.53f}{2} = 0.49f.$$

Noise:

$$\frac{6.2\mu}{2} = 3.1\mu$$

CT in:

$$\frac{0.21f + 0.12f}{2} = 0.165f.$$

CT out:

$$\frac{5.1n + 5.1n}{2} = 5.1n.$$

(P)

delay: $\frac{0.45f + 0.58f}{2} = 0.51f$

Noise: $\frac{6.2\mu}{2} = 3.1\mu$

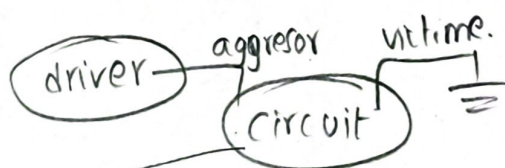
CT in =

$$\frac{0.21f + 0.12f}{2} = 0.165f.$$

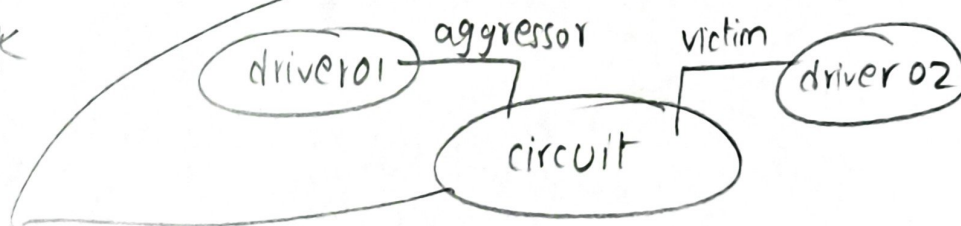
CT out = $\frac{5.1n + 5.1n}{2} = 5.1n.$

S.No	Model	P-Delay	Noise	CT-inphase	CT-outphase.
01.	Defect free	0.49f	3.1 μ	0.12f	5.1n.
02.	Defected CG (Chao liu)	0.55f	3.07 μ	0.165f	5.1n.
03.	Defected CG (P)	0.545f	3.07 μ	0.17f	5.1n.
04.	Defected RL (N)	0.49f	3.1 μ	0.165f	5.1n.
05.	Defected RL (P)	0.51f	3.1 μ	0.165f	5.1n.

noise - delay



crosstalk



circuit

defect free

defected cg

defected RL

values change

defected cg

defected RL

cgC

cgP

RLN

RLP

R_u, C_a
change.